

Local Setup

Follow the steps below to set up the Furniture E-Commerce Microservices project on your local machine.

Prerequisites

Install the following software before running the project:

Software	Version
Visual Studio	2022/2026
.NET SDK	.NET 8
SQL Server	2022/2025
SQL Server Management Studio (SSMS)	20/21/22
Docker Desktop	Latest
Git	Latest
RabbitMQ	Docker Image
Node.js	Latest LTS
Angular CLI	v22

Step 1 - Clone the Backend Repository

```
```bash
git clone https://github.com/pravinnarajagopal-star/ECommerce_Furniture.git
```
```

Open the solution:

```
```
ECommerce_Furniture.sln
```
```

using Visual Studio.

Step 2 - Create the Databases

Create the following databases in SQL Server:

```
...
CustomerDB
ProductDB
OrderDB
PaymentDB
NotificationDB
...
```

Step 3 - Configure Connection Strings

Update the **appsettings.json** file in each microservice.

Example:

```
```json
"ConnectionStrings": {
 "DefaultConnection": "Server=localhost;Database=ProductDB;User
Id=sa;Password=YourPassword;TrustServerCertificate=True"
}
...
```

> **Note:** Update the database name for each microservice accordingly.

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### ## Step 4 - Create Database Tables

Choose **one** of the following approaches:

#### ### Option 1 - Entity Framework Core (Recommended)

Open **Package Manager Console**.

Select the appropriate startup project and execute:

```
```powershell
Update-Database
...
```

Repeat this step for each microservice.

Option 2 - Execute SQL Scripts

If you prefer not to use Entity Framework migrations, execute the SQL scripts provided in the ****Database Design**** folder.

The scripts include:

- Database creation
- Table creation
- Primary and Foreign Keys
- Sample Master Data
- Seed Data

Step 5 - Start RabbitMQ Using Docker

Run the following command:

```
```bash
docker run -d ^
--hostname rabbitmq ^
--name rabbitmq ^
-p 5672:5672 ^
-p 15672:15672 ^
rabbitmq:3-management
```
```

RabbitMQ Management Portal:

...

<http://localhost:15672>

...

Default Credentials:

| Username | Password |
|----------|----------|
| ----- | ----- |
| guest | guest |

Step 6 - Run the Microservices

Start the following projects:

- ApiGatewayYarp
- CustomerAuthService
- ProductService
- OrderService
- PaymentService
- NotificationService

Ensure all services are running successfully before launching the Angular application.

Step 7 - Verify Swagger Endpoints

Open the following Swagger URLs:

| Service | URL |
|---------------------|--------------------------------|
| ----- | ----- |
| API Gateway | https://localhost:7000/swagger |
| CustomerAuthService | https://localhost:7001/swagger |
| ProductService | https://localhost:7002/swagger |
| OrderService | https://localhost:7003/swagger |
| PaymentService | https://localhost:7004/swagger |
| NotificationService | https://localhost:7005/swagger |

Angular UI Setup

Clone the Angular repository:

```
```bash
git clone https://github.com/pravinnarajagopal-star/FurnitureECommerce-AngularUI.git
```
```

Navigate to the project folder:

```
```bash
cd FurnitureECommerce-AngularUI
```
```

Install dependencies:

```
```bash
npm install
```
```

Run the Angular application:

```
```bash
ng serve
```
```

Open your browser:

```
```
http://localhost:4200
```
```

Docker Deployment

Build the Docker images:

```
```bash
docker compose build
```
```

Start all containers:

```
```bash
docker compose up -d
```
```

Verify the running containers:

```
```bash
docker ps
```
```

Stop all containers:

```
```bash
docker compose down
```
```

Application Flow

...

Angular 22 UI



YARP API Gateway



Customer Authentication Service



JWT Authentication



Product Service



Order Service



RabbitMQ



Payment Service



RabbitMQ



Notification Service

...

Future Enhancements

- Redis Cache Integration
- Serilog Logging
- Health Checks
- Centralized Exception Handling
- Unit Testing (xUnit)

- Integration Testing
- GitHub Actions CI/CD
- Azure Deployment
- Kubernetes (AKS)
- Monitoring with Prometheus & Grafana

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